Four Pillars of Health Research:

1. Biomedical Research: goal of understanding **normal and abnormal human functioning** at molecular, cellular, organ system, and whole body levels (includes development of tools/techniques applied for this purpose)

- Focus on developing **new therapies/devices** that improve health/quality of life for individuals up to point where they are tested on human subjects
- 2 main categories: **fundamental/basic research, applied research**
- Leads to new ways of identifying/diagnosing disease, interventions to prevent illness, tools/equipment to improve care/outcomes, medicines/vaccines/therapies
- Often involves experts from a wide range of fields to answer big questions through research studies, lab experiments, tests of new treatments, etc

2. Clinical Research: collects evidence needed to improve clinical practice

- Addresses a wide range of challenges related to improving the prevention, diagnosis, and treatment of disease and illness
- 2 main categories:
    1. **Observation studies**: questionnaires and medical tests to analyze data
    2. **Clinical trials**: testing new inventions

3. Health Services Research: improved healthcare (healthcare system)

- Study how health care services are organised, supported, delivered across Canada to improve overall system
- Generates info needed to help **enhance care for patients**, reduce costs, address needs of healthcare providers
- *improve efficiency and effectiveness by enhancing quality of care, improving health care provider experience, reducing costs by maximizing value of care, improving overall health of pop*
- Collecting and analyzing health data, interviewing providers and patients, running studies, examining successful healthcare systems/services

4. Social, cultural, environmental, and population health research:

- Research uncovers ways that social, cultural, environmental, occupational, and economic factors can affect health (for better or for worse)
- Conducting focus groups with stakeholders, evaluating national/international policies, analyzing health data and trends over time, designing health interventions and programs